✦ MACHINE LEARNING ✦

The Science of Learning from Data

> "Experience is not what happens to you. It's what you do with the data you collect."
> – Unknown"

1. WHAT IS MACHINE LEARNING?

ML is a subset of Artificial Intelligence that enables systems to learn patterns from data and make decisions or predictions without being explicitly programmed.

DATA → LEARN (PATTERNS) → PREDICT / DECIDE → IMPROVE (OVER TIME)

Goal: Build models that generalize well to unseen data.

2. WHY MACHINE LEARNING?

- ★ Learns from data and improves with experience.
- ★ Handles complex patterns humans can't write as rules.
- ★ Automates predictions and decisions.
- ★ Works across industries and real-world problems.
- ★ Drives AI applications we use every day!

SPAM DETECTION | RECOMMENDER SYSTEMS | HEALTHCARE DIAGNOSIS | SALES FORECASTING | SELF DRIVING CARS

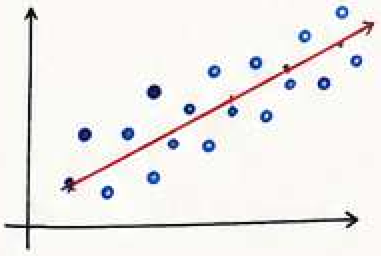
3. TYPES OF MACHINE LEARNING

SUPERVISED LEARNING

Learn from labeled data (input-output pairs).

Examples:

- Classification
- Regression



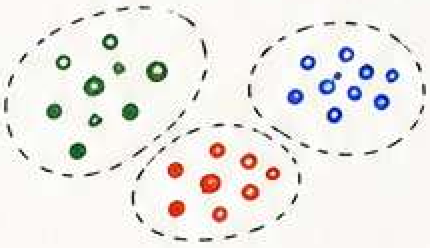
Use case: Predict house prices, email spam, disease diagnosis.

UNSUPERVISED LEARNING

Find patterns/structure in unlabeled data.

Examples:

- Clustering
- Dimensionality Reduction
- Association Rule Learning



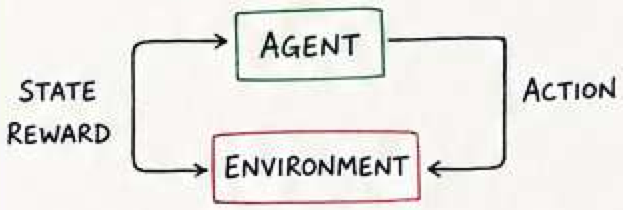
Use case: Customer segmentation, market basket analysis.

REINFORCEMENT LEARNING

Learn by interacting with an environment and maximizing rewards.

Examples:

- Q-Learning
- Deep Q-Networks
- Policy Gradients



Use case: Robotics, game playing, autonomous systems.

4. WORKFLOW OF A MACHINE LEARNING PROJECT

1. Problem Definition → 2. Data Collection → 3. Data Preprocessing → 4. Feature Engineering → 5. Model Training → 6. Evaluation → 7. Deployment & Monitoring

5. IMPORTANT TERMS

- **Dataset**: Collection of data samples.
- **Feature**: An input variable used for prediction.
- **Target (Label)**: The output variable to predict.
- **Training Data**: Used to train the model.
- **Test Data**: Used to evaluate the model.
- **Overfitting**: Model performs well on training data but poorly on unseen data.
- **Underfitting**: Model is too simple to learn patterns in data.
- **Generalization**: Model performs well on unseen data.

☆ Remember: Better Data → Better Features → Better Model → Better Decisions

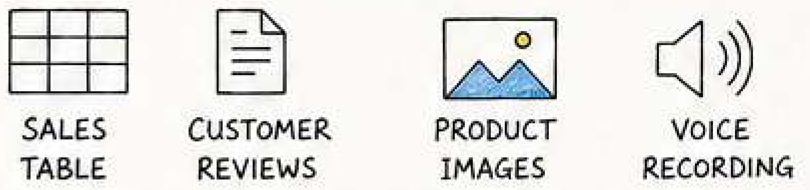
6. DATA : THE FOUNDATION OF ML

> "Garbage In, Garbage Out." – Quality data leads to quality models.

6.1 TYPES OF DATA

- Structured Data : Tables (rows & columns)
- Unstructured Data : Text, images, audio, video
- Semi-structured Data : JSON, XML, logs

EXAMPLES

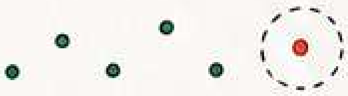


6.2 COMMON DATA ISSUES

✗ Missing values

23	NA	45

✗ Outliers

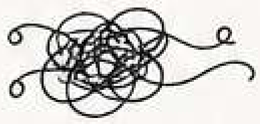


✗ Inconsistent formats "M", "Male", "m"

✗ Duplicate records

ID : 101, Name : John
ID : 101, Name : John

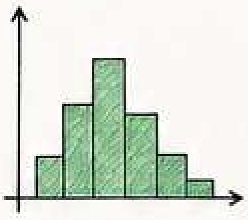
✗ Irrelevant / Noisy data



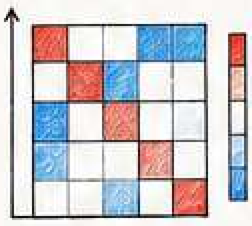
7. EXPLORATORY DATA ANALYSIS (EDA)

- Understand data before building models.
- Find patterns, anomalies and insights.
- Helps in better feature engineering.

DISTRIBUTION



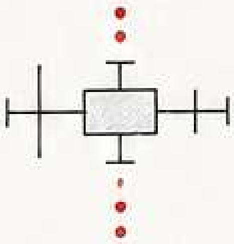
CORRELATION



MISSING VALUES

A	B	C
✓	✓	✓
✓	NA	NA
NA	✓	✓

OUTLIERS



8. FEATURE ENGINEERING

- Convert raw data into meaningful features.
- Good features > Complex Models.
- Improves model performance significantly.

TECHNIQUES

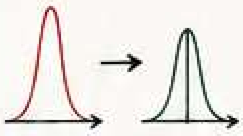
ENCODING
(Categorical → Numeric)

Color
Red
Blue
Green

→

1
1
0
2

SCALING
(Bring to same scale)



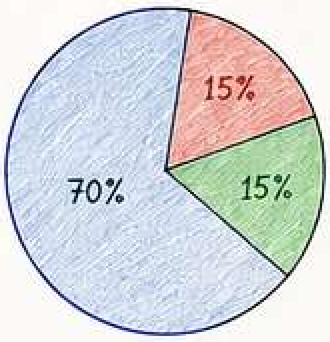
FEATURE CREATION
(New from existing)

Date → Day, Month, Weekday

FEATURE SELECTION
(Keep useful, remove noise)

✓	✗
✓	✗
✓	✓
✓	✗

9. TRAIN – VALIDATION – TEST SPLIT



🟦 TRAINING SET (70%)
→ Used to train the model

🟩 VALIDATION SET (15%)
→ Used for tuning hyperparameters and model selection

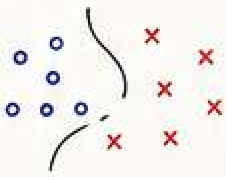
🟥 TEST SET (15%)
→ Used to evaluate final model performance

TIPS

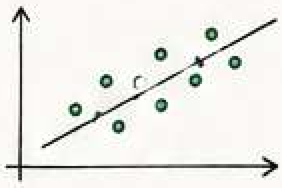
- Never use test data for training.
- Keep test set completely unseen.
- For small datasets → Use Cross-Validation.

10. COMMON TASKS IN MACHINE LEARNING

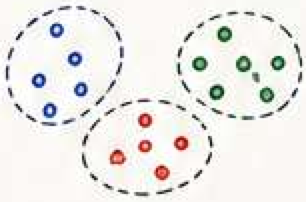
CLASSIFICATION
Predict a category (Yes/No, Spam/Ham)



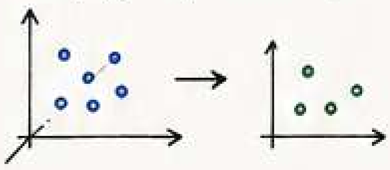
REGRESSION
Predict a continuous value (price, salary)



CLUSTERING
Group similar data points



DIMENSIONALITY REDUCTION
Reduce features while keeping important info



ASSOCIATION RULE LEARNING
Find interesting relationships

{Bread, Butter} → Milk

> Key Takeaway : Understand your data deeply. Good data + Good features = Great Models.

11. BUILDING A MACHINE LEARNING MODEL

1. Define Problem → 2. Collect Data → 3. Prepare Data → 4. Train Model → 5. Evaluate Model → 6. Deploy & Monitor

1. Define Problem	2. Collect Data	3. Prepare Data	4. Train Model	5. Evaluate Model	6. Deploy & Monitor
What do we want to predict?	Gather relevant data from sources.	Clean, transform and engineer features.	Choose algorithm and train on training data.	Test performance using metrics and validation.	Use model in real world and track performance.

> Remember:
> "A good model is not just accurate, but useful, reliable and used."

12. EVALUATION METRICS

12.1 CLASSIFICATION METRICS

		Actual	
		Positive	Negative
Predicted	Positive	TP (True Positive)	FP (False Positive)
	Negative	FN (False Negative)	TN (True Negative)

- Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$
- Precision = $\frac{TP}{TP + FP}$
- Recall (Sensitivity) = $\frac{TP}{TP + FN}$
- F1 Score = $2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$
- Specificity = $\frac{TN}{TN + FP}$

12.2 REGRESSION METRICS

- MAE (Mean Absolute Error)
  $= \frac{1}{n} \sum |y_i - \hat{y}_i|$
- MSE (Mean Squared Error)
  $= \frac{1}{n} \sum (y_i - \hat{y}_i)^2$
- RMSE = $\sqrt{MSE}$
- R^2 Score = $1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$

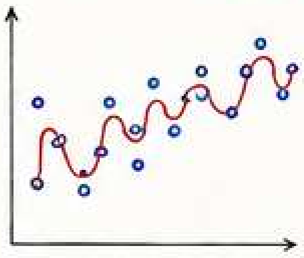
R^2 SCORE GUIDE

- 1 → Perfect fit
- 0 → No better than mean
- < 0 → Worse than mean prediction

Choose the right metric based on the problem and business goal!

13. OVERFITTING vs UNDERFITTING

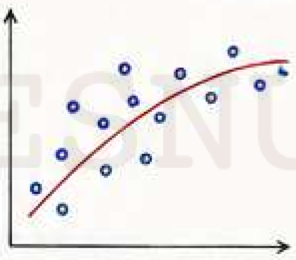
OVERFITTING



- Model learns noise in training data.
- High variance
- Performs well on training data but poorly on test data.

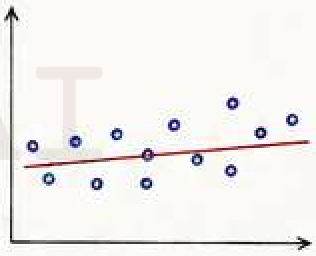
Solution: Get more data, simplify model, use regularization, cross-validation.

GOOD FIT



- Captures pattern well.
- Good balance of bias and variance.
- Performs well on both training and test data.

UNDERFITTING



- Model too simple to learn pattern.
- High bias
- Performs poorly on both training and test data.

Solution: Use more complex model, add features, train longer.

14. HANDLING COMMON DATA PROBLEMS

MISSING VALUES

- Remove rows/columns (if less missing)
- Fill with mean/median (numerical)
- Fill with mode (categorical)
- Use KNN Imputer

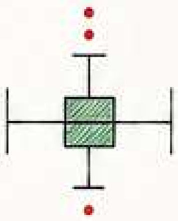
A	B	C
1	2	NA
4	NA	6
7	8	9

→

A	B	C
1	2	5
4	3	6
7	8	9

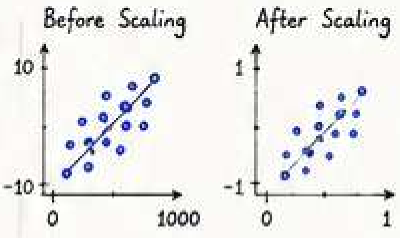
OUTLIERS

- Detect using boxplot or Z-score.
- Handle by capping, transforming or removing.



SCALING

- StandardScaler (mean = 0, std = 1)
- MinMaxScaler (range = 0 to 1)



CATEGORICAL DATA

- Nominal → One-Hot Encoding
- Ordinal → Label Encoding or Ordinal Encoding

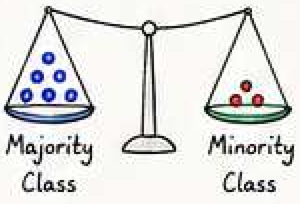
Color
Red
Blue
Green

→

Red	Blue	Green
1	0	0
0	1	0
0	0	1

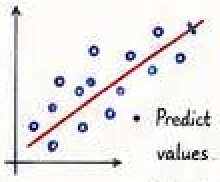
IMBALANCED DATA

- Use class_weight
- Oversampling (SMOTE)
- Undersampling
- Choose right metrics (Recall, F1)



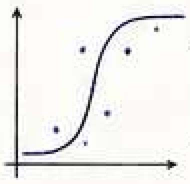
15. POPULAR ALGORITHMS IN SCIKIT-LEARN

LINEAR REGRESSION



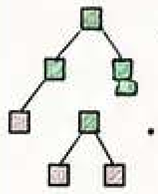
- Predict continuous values.
- Simple and interpretable.

LOGISTIC REGRESSION



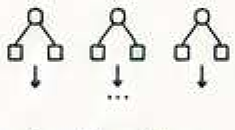
- Classification (0 or 1).
- Outputs probabilities.

DECISION TREE



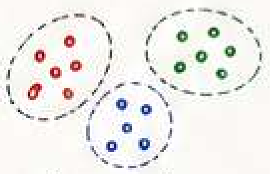
- Easy to interpret.
- Can handle non-linear data.

RANDOM FOREST



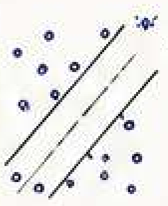
- Ensemble of trees.
- More accurate and robust.

K-MEANS CLUSTERING



- Unsupervised.
- Groups data into K clusters.

SVM (SUPPORT VECTOR MACHINE)



- Finds best boundary between classes.

Key Takeaway: Good data + Right algorithm + Proper evaluation = Powerful ML model!

16. HYPERPARAMETER TUNING

Hyperparameters are settings that are set before training the model.

Examples

- Learning rate (η)
- Max depth (Tree)
- C (SVM, Logistic Reg.)
- n_estimators (Random Forest)
- k (KNN)

Tuning Methods

- Grid Search : Try all combinations in a given range.
- Random Search : Try random combinations.
- Bayesian Optimization : Smart search using past results.

Goal : Find the best combination for better performance and generalization.

☆ Tuning improves performance, but remember : A perfectly tuned model on the wrong data is still the wrong model.

17. CROSS-VALIDATION

- Helps to evaluate model performance more reliably.
- Reduces overfitting to a single train-test split.
- Common methods :
  1. K-Fold Cross-Validation
  2. Stratified K-Fold (for imbalanced data)
  3. Leave-One-Out (LOOCV)

K-FOLD CROSS-VALIDATION (K = 5)

Iteration 1	Test	Train	Train	Train	Train
Iteration 2	Train	Test	Train	Train	Train
Iteration 3	Train	Train	Test	Train	Train
Iteration 4	Train	Train	Train	Test	Train
Iteration 5	Train	Train	Train	Train	Test

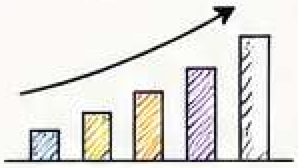
Final Score = Average of all 5 test scores

Why ?

✓ More stable estimate of performance

✓ Uses all data for training and testing

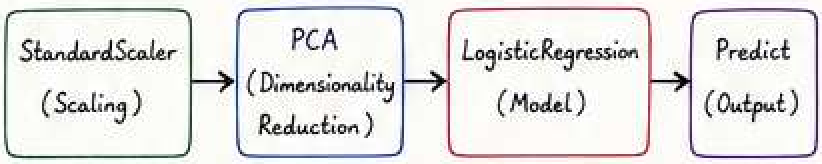
✓ Better for small datasets



18. PIPELINES IN SCIKIT-LEARN

- Chains multiple steps (preprocessing + model) into a single workflow.
- Prevents data leakage.
- Makes code clean and reproducible.

Example Pipeline :



```
# Example in Scikit-learn
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression

pipeline = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('pca', PCA(n_components=5)),
    ('model', LogisticRegression())
])
pipeline.fit(X_train, y_train)
predictions = pipeline.predict(X_test)
```

Benefits

- Reproducible
- Modular
- Cleaner Code
- Better Generalization

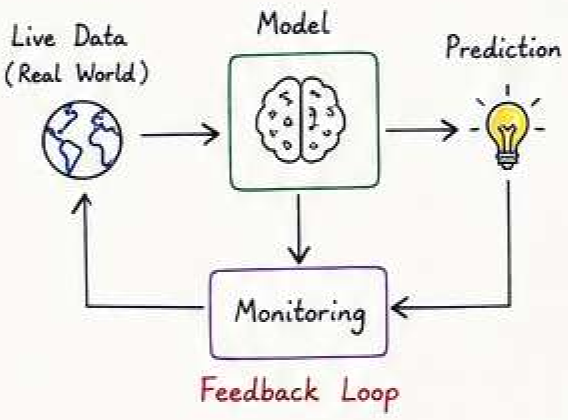
19. MODEL DEPLOYMENT & MONITORING

Deployment Options

- Cloud Platforms (AWS, GCP, Azure)
- Web APIs (Flask, FastAPI)
- Containers (Docker)
- Edge / Mobile Devices
- Batch Predictions

Monitoring is Important!

- Track performance over time.
- Detect data drift.
- Detect concept drift.
- Retrain or update the model when needed.



20. KEY TAKEAWAYS

☑ Understand the problem deeply.
☑ Explore and clean the data.
☑ Engineer good features.
☑ Choose the right model.
☑ Evaluate with the right metrics.

☑ Prevent overfitting.
☑ Handle data problems smartly.
☑ Tune hyperparameters.
☑ Use cross-validation.
☑ Deploy, monitor and improve continuously.

Machine Learning is an iterative process. There is no "best algorithm" – only the best approach for the problem and data you have.

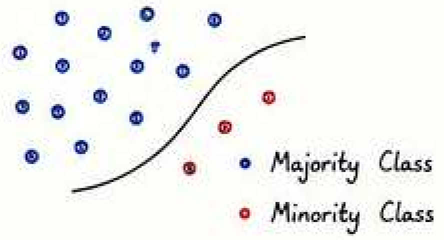
Think. Experiment. Learn. Improve.

Data is the fuel. Algorithms are the engine. But curiosity is the driver.

21. HANDLING IMBALANCED DATA

When one class is much more frequent than the other.

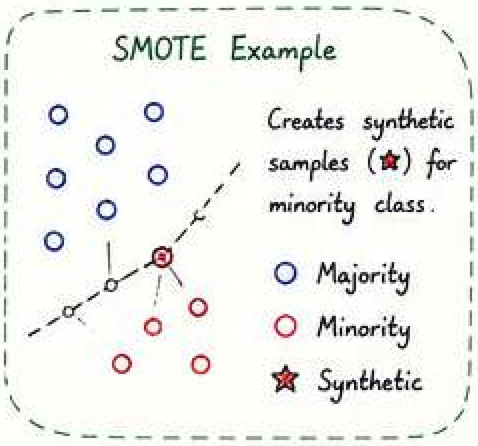
Example :



Techniques

- Resampling Methods
  - Oversampling (SMOTE)
  - Undersampling
- Use class_weight='balanced'
- Choose appropriate metrics (Recall, F1 Score, PR-AUC)
- Collect more data

SMOTE Example



Remember :

Accuracy can be misleading.

Focus on the metric that matters for your problem!

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22. MODEL VALIDATION TECHNIQUES

1. Holdout Validation

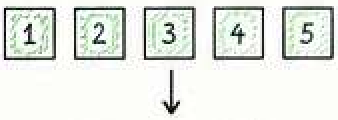
Split data into train and test sets.

Simple and fast.

But variance can be high.

2. K-Fold Cross-Validation

Split data into K parts.



Train on K-1 folds

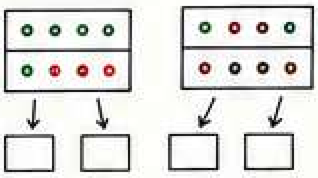
Test on 1 fold

Repeat K times.

More stable estimate.

3. Stratified K-Fold

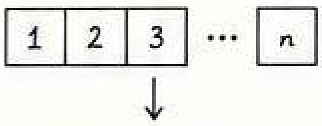
Maintains class distribution in each fold.



Best for imbalanced data.

4. Leave-One-Out (LOOCV)

Use one sample as test data, rest for training.



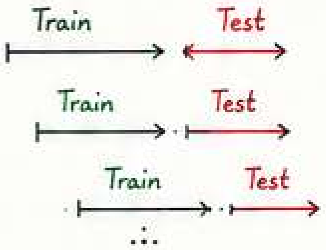
Repeat n times.

Low bias, high variance.

Computationally expensive.

5. Time Series Split

Used for sequential data.



Preserves time order.

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23. PRACTICAL TIPS FOR BETTER MODELS

☑ Start simple, then increase complexity.
☑ Visualize data and relationships.
☑ Handle missing values properly.
☑ Detect and treat outliers.
☑ Use pipelines to avoid data leakage.
☑ Validate your model, don't just train it.
☑ Understand your business problem deeply.
☑ Iterate, experiment and learn! ☺

Questions to Ask

- What is the goal?
- What data do I have?
- What could go wrong?
- Is my evaluation metric aligned with the goal?
- How will this model be used in the real world?

Golden Rule

There is no perfect model, only the right model for the right problem.

Common Pitfalls

✗ Using accuracy for imbalanced data
✗ Data leakage
✗ Overfitting
✗ Ignoring business context
✗ Not validating properly

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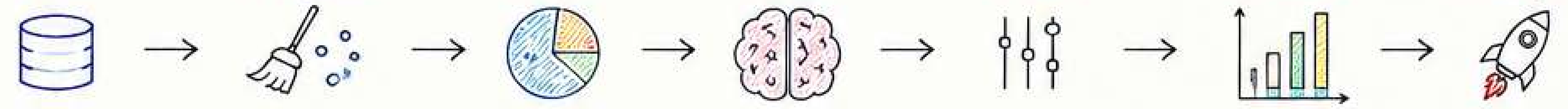
24. END-TO-END EXAMPLE (CLASSIFICATION)



1. Data → Collect raw data
2. Preprocess → Clean, handle missing values, encode categorical features
3. Split → Train (70%) Validation (15%) Test (15%)
4. Train → Choose algorithm and train on training data
5. Tune → Tune hyperparameters using cross-validation on validation set
6. Evaluate → Evaluate on test set using appropriate metrics
7. Deploy → Deploy model and monitor in real world

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25. SUMMARY CHEAT SHEET

Goal : Define problem clearly.
Data : Gather, clean and understand.
Features : Engineer meaningful features.
Model : Choose right algorithm.
Evaluate : Use right metrics.
Tune : Optimize hyperparameters.
Deploy : Put into production.
Monitor : Track and improve.

Problem Type	Algorithm Ideas	Key Metrics	Notes
Classification (2 classes)	Logistic Regression, SVM, Random Forest, XGBoost	Accuracy, Precision, Recall, F1, ROC-AUC	Use ROC-AUC for imbalanced data.
Classification (Multi-class)	Random Forest, SVM, XGBoost, Neural Nets	Accuracy, F1 (macro), Confusion Matrix	Check per-class performance.
Regression	Linear Regression, Ridge, Random Forest, XGBoost	MAE, MSE, RMSE, R^2 Score	Look at residuals too.
Clustering	K-Means, DBSCAN, Hierarchical	Silhouette Score, Davies-Bouldin Index	No labels available.
Association Rule Learning	Apriori, FP-Growth	Support, Confidence, Lift	Market basket analysis.

Learn the concepts. Practice on real data. Build projects. That's how you become a ML Engineer!